

Announced Rescue Pricing with Latent Market Thickness

Group-meeting note — benchmark, numerical pattern, and formal model

27 August 2026

Preview. A publicly known single driver gives an exact cutoff-WPBE and a closed-form optimal menu. With latent Poisson supply, the completion value of rescue pricing is small in thin markets, largest at an intermediate thickness, and small again in thick markets. For $(\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0)$, the deterministic numerical solution peaks near $m = 9.72$ and improves optimized completion by 6.16 percentage points. The endpoint pattern is proved; strict single-peakedness is a conjecture.

1 Exact one-rider/one-driver benchmark

1.1 Closed-form solution

The rider has $v \sim U[0, 1]$, the driver has $c \sim U[0, 1]$, and the platform maximizes completion. Delayed rider value is βv and delayed incumbent surplus is $\delta(p_j - c)$. The platform commits to $0 \leq p_1 \leq p_2 \leq 1$. The number of drivers is publicly known to be exactly one; there is no survival shock or fresh entry.

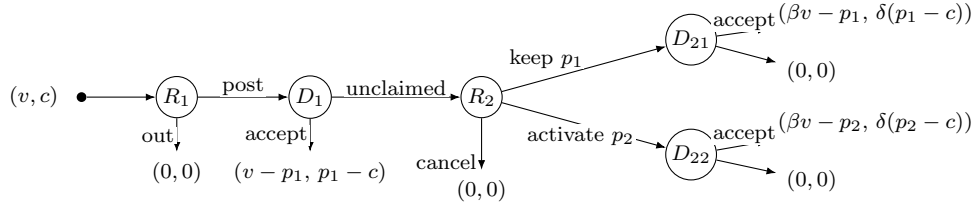


Figure 1: Extensive form.

For an active menu $0 \leq p_1 < p_2 < \beta$, the driver accepts immediately iff $c \leq \hat{c}$. After rejection, the rider cancels, repeats p_1 , or activates p_2 according to

$$v < \frac{p_1}{\beta}, \quad \frac{p_1}{\beta} \leq v \leq v^M(\hat{c}), \quad v > v^M(\hat{c}), \quad v^M(\hat{c}) = \frac{p_1 + p_2 - \hat{c}}{\beta}.$$

The marginal driver's accept-versus-wait indifference gives the WPBE cutoff

$$\hat{c}(p_1, p_2) = \left[p_1 - \frac{\delta(p_2 - p_1)(\beta - p_2)}{\beta(1 - p_1) - \delta(\beta - p_2)} \right]_+, \quad (1)$$

and equilibrium completion is

$$M_1(p_1, p_2) = (1 - p_1)\hat{c} + \frac{(\beta - p_2)(p_2 - \hat{c})}{\beta}. \quad (2)$$

Thus the cutoff is calculated from the WPBE; the platform then optimizes over menus subject to this equilibrium response.

Fixing a target cutoff reduces the two-price problem to one dimension. The unique continuation envelope and its implementing first price are

$$\hat{p}_2(\hat{c}) = \frac{\beta + \hat{c}}{2}, \quad \hat{p}_1(\hat{c}) = \frac{1 + \hat{c} - \sqrt{\Delta(\hat{c})}}{2}, \quad \Delta(\hat{c}) = (1 - \hat{c})^2 - \frac{\delta}{\beta}(\beta - \hat{c})^2. \quad (3)$$

Substitution into (2) gives

$$M_1^* = \max \left\{ \frac{1}{4}, \max_{\hat{c} \in [0, \beta]} J_\delta(\hat{c}) \right\}, \quad J_\delta(\hat{c}) = \frac{\hat{c}[1 - \hat{c} + \sqrt{\Delta(\hat{c})}]}{2} + \frac{(\beta - \hat{c})^2}{4\beta}. \quad (4)$$

If $\beta \leq 1/2$ or $\delta = 1$, the optimal value is the flat benchmark $1/4$. If $\beta > 1/2$ and $\delta < 1$, J_δ is strictly concave and has a unique interior maximizer, which implements $p_1^* < p_2^*$.

1.2 Numerical illustration

At $\beta = \delta = 4/5$, the first-order condition becomes $12 - 20\hat{c} = 5\hat{c}\sqrt{9 - 10\hat{c}}$. Squaring yields $(5\hat{c} - 4)(50\hat{c}^2 + 75\hat{c} - 36) = 0$; checking the unsquared equation and feasibility leaves the positive root of the quadratic. Its discriminant is $75^2 + 4(50)(36) = 225 \cdot 57$, which explains the appearance of $\sqrt{57}$:

$$\hat{c}^* = \frac{3(\sqrt{57} - 5)}{20} = .382475, \quad p_1^* = \frac{11 + \sqrt{57}}{40} = .463746, \quad p_2^* = \frac{1 + 3\sqrt{57}}{40} = .591238.$$

The optimal rescue menu completes 25.9581% of requests, compared with 25.0000% under the best flat price.

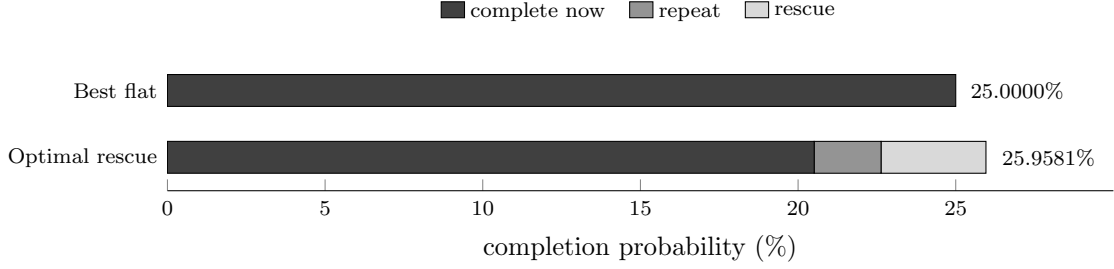


Figure 2: Exact-one-driver completion at $\beta = \delta = .8$. The 0.9581 pp gain combines less completion now with repeat and rescue completion later.

β	δ	$\hat{c}^*$	p_1^*	p_2^*	M_1^*
.6	.5	.460069	.467733	.530034	.253038
.6	.8	.478613	.488210	.539306	.251089
.8	.5	.370421	.420148	.585210	.272458
.8	.8	.382475	.463746	.591238	.259581
.9	.5	.329857	.405843	.614928	.286282
.9	.8	.314018	.453693	.607009	.266932

Each row re-solves the WPBE and optimal menu. More patient riders support a larger rescue region; less patient incumbents make waiting less attractive and increase the completion value of escalation.

2 Three new directions share one nested WPBE design

Takeaway. Starting from the two-page benchmark, we change only second-window supply: retained incumbents, time-homogeneous fresh arrivals, and finally an expanded rescue footprint. Prices and search are always optimized outside the induced cutoff-WPBE.

Architecture	Policy	Who can volunteer in window 2	What it identifies
I: retained incumbents	(p_1, p_2)	First-window rejectors who remain available	Same-order recall value
A: fixed footprint + arrivals	(p_1, p_2)	Retained rejectors plus an independent fresh core cohort	Time-homogeneous replenishment
E: expanded rescue search	(p_1, p_2, s)	The same core supply plus an outer annulus reached only in rescue	Incremental geographic reach

Here p_j is one anonymous transaction price: the rider pays p_j and the assigned driver receives p_j . The local per-window flows are

$$N_1 \sim \text{Pois}(m), \quad N_2^C \sim \text{Pois}(m), \quad N_2^O(s) \sim \text{Pois}(m(s-1)). \quad (5)$$

Conditional on first-window cutoff a , define retained-incumbent, fresh-core, and outer-annulus volunteer intensities by

$$I_j(a) = \omega m[F(p_j) - F(a)]_+, \quad E_j^C = mF(p_j), \quad E_2^O(s) = m \int_1^s F(p_2 - \tau(\sqrt{u} - 1)) du. \quad (6)$$

The total terminal intensity is

$$\lambda_j^r(a) = I_j(a) + \mathbf{1}\{r \neq \text{I}\} E_j^C + \mathbf{1}\{r = \text{E}, j = 2\} E_2^O(s), \quad C_j^r = 1 - e^{-\lambda_j^r}, \quad h(\lambda) = \frac{1 - e^{-\lambda}}{\lambda}. \quad (7)$$

Thus every λ is an expected Poisson volunteer count, C is coverage, and h is one volunteer's lottery win probability. Only $\text{A} \subset \text{E}$: $s = 1$ reproduces the fixed footprint, while the rescue radius is $R_0\sqrt{s}$.

Information and incentives. Incumbents have seen the order and commitment, so they may reject and wait. Fresh drivers first see it in window 2. Universal rejection is public; the rider observes failure but not realized counts, and uses equilibrium coverage. A rejector remains eligible with probability ω .

Two different costs. An assigned outer driver pays the post-assignment pickup cost embedded in (6). Separately, after failure the platform reserves capacity for its promised outer footprint:

$$Q^{\text{com}} = (1 - p_1)e^{-ma}m(s-1). \quad (8)$$

Thus free search is not built into the objective.

Every plotted policy is evaluated at its own cutoff-WPBE:

$$V_r(\theta) = \max_{\mu_r} J_r(\mu_r, a^*(\mu_r; \theta)), \quad J_r = M_r - \kappa Q_r^{\text{com}}, \quad \boxed{\text{fix } p_1 \rightarrow \max_{p_2, s} \rightarrow \max_{p_1}}. \quad (9)$$

The inner cutoff a^* , rider continuation, and supply composition are all re-solved at every outer-policy evaluation; s is present only for E.

3 Directions 1–2: isolate the value of fresh arrivals

Takeaway. Adding a fresh core cohort has its largest displayed value in a moderately thick market; the option to search outside the core is most valuable earlier, when the local market is still thin.

Fresh replenishment matters later; geographic reach matters first

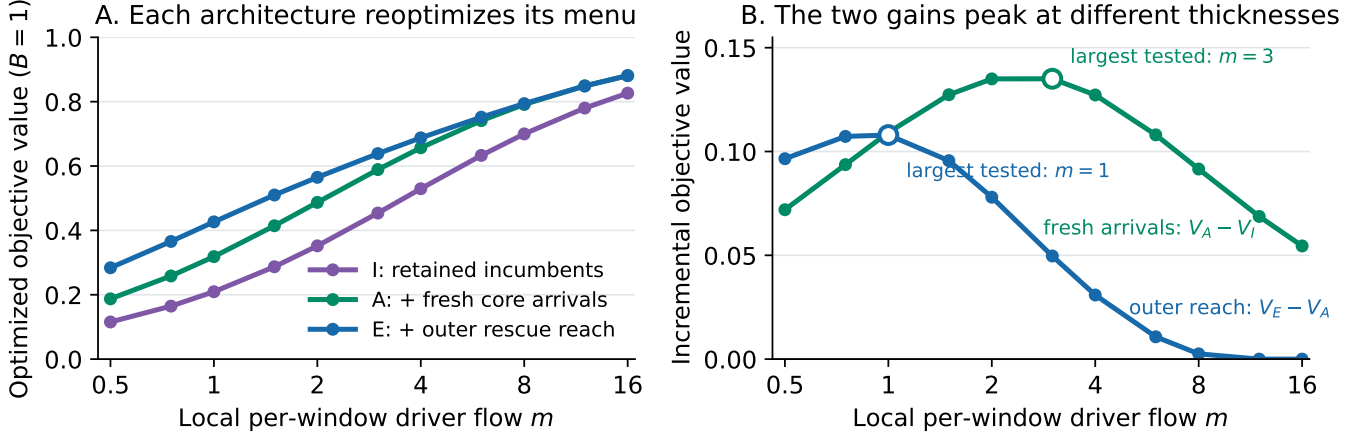


Figure 1: Illustrative slice $(\beta, \delta, \omega, \tau, \kappa, \bar{s}) = (.8, .8, .8, .25, .0125, 4)$ with $v, c \sim U[0, 1]$ and $B = 1$. Every point separately reoptimizes all controls available to that architecture and re-solves the cutoff-WPBE. I and A report completion; E reports completion net of committed-reach cost.

- **I $\rightarrow$ A: replenishment.** Fresh arrivals directly raise second-window coverage and also compete with rejectors who wait. The gain is therefore an equilibrium response, not a fixed-policy supply addition.
- **A $\rightarrow$ E: reach option.** Because $s = 1$ is feasible, this net option value is nonnegative. Its largest tested value occurs around $m = 1$, before the fresh-core value reaches its largest tested value around $m = 3$.
- **Interpret the peaks cautiously.** The plot supports two distinct economic regions and an ordered-peak conjecture; it does not claim unique global peaks in the fully endogenous WPBE model.

4 Direction 3: expanded search is an early-thickness tool

Takeaway. Market thickness changes the mix of recovery tools. Search scope contracts sharply with m , but a positive rescue price increment remains even after the optimized footprint returns to the core.

Search scope contracts before the rescue price increment disappears

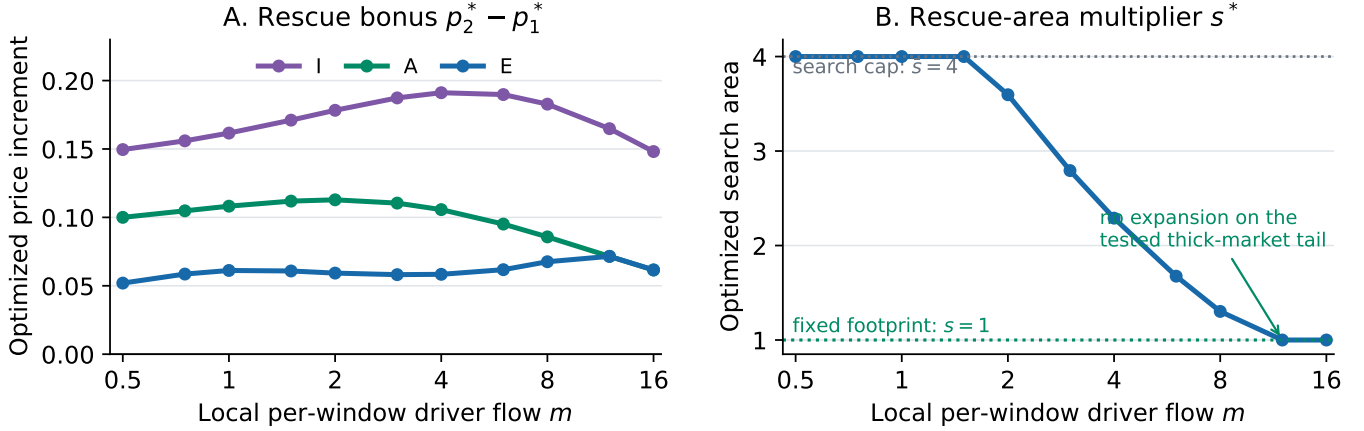


Figure 2: Same calibration and equilibrium-constrained optimization as page 4. The right panel uses committed-reach cost; $s = 1$ means no expansion beyond the fresh core cohort.

Market region	Main shortage	Numerical mechanism intuition
Thin	Too few reachable drivers	Expand aggressively; the tested optimum reaches the search cap.
Intermediate	Coverage and strategic waiting both matter	Combine a rescue increment with a gradually shrinking outer footprint.
Thick	The core is already reliable	Set $s = 1$ on the tested tail, while retaining a positive $p_2 - p_1$.

Current story. Fresh arrivals and expanded search are different sources of recovery value. Search is an early-thickness tool; rescue pricing continues to manage screening and delayed acceptance after geographic expansion is no longer worth its capacity cost.

The next theory step is deliberately deferred: primitive conditions for peak ordering and the exact no-search threshold in κ . The complete theory note keeps those claims, proofs, counterexamples, and literature boundaries separate from this five-page intuition-and-simulation presentation.